IN SILICO EVALUATION OF EUGENOL AND CURCUMIN AS DIRECT INHA INHIBITORS IN *MYCOBACTERIUM TUBERCULOSIS*

A comparative molecular docking and ADMET study against isoniazid: Oct 19, 2026

Study details

Field	Entry
Author	[Name, enrolment number]
Project Type	Independent Computational Research / Self-Study
Target organism	*Mycobacterium tuberculosis* H37Rv
Protein target	Enoyl-[acyl-carrier-protein] reductase (InhA), EC 1.3.1.9
Crystal structure	PDB 2B35, 2.30 Å, InhA inhibited by triclosan
UniProt accession	P9WGR1 (INHA_MYCTU)
Software	SwissADME; PubChem; Mcule 1-Click Docking (AutoDock Vina 1.1.2)
Analysis performed	19/10/2026

ABSTRACT

Eugenol and curcumin both dock into the InhA substrate-binding site more favourably than isoniazid by raw score, but only eugenol holds that advantage once the score is corrected for molecular size. Isoniazid is a prodrug: it must be oxidised by the mycobacterial catalase-peroxidase KatG before the resulting adduct can inhibit enoyl-ACP reductase (InhA), and loss-of-function mutations in *katG* account for most clinical isoniazid resistance. Compounds that engage InhA without this activation step are therefore of continuing interest. Two dietary phenolics, eugenol (*Syzygium aromaticum*) and curcumin (*Curcuma longa*), were profiled in SwissADME and docked into the triclosan-bound InhA structure (PDB

2B35) with AutoDock Vina, using isoniazid as reference. All three satisfied Lipinski's rule of five with no violation and were predicted to show high gastrointestinal absorption. Best-pose scores were −8.3 kcal/mol for curcumin, −7.4 for eugenol and −6.4 for isoniazid. Normalised per heavy atom, however, eugenol (0.62) and isoniazid (0.64) are comparable while curcumin falls to 0.31, indicating that curcumin's apparent advantage is largely a consequence of its size. Curcumin additionally triggers a pan-assay interference alert and is known to be chemically unstable and poorly bioavailable. On the evidence presented here, eugenol is the better-supported scaffold for experimental follow-up.

Keywords: InhA; enoyl-ACP reductase; molecular docking; eugenol; curcumin; ligand efficiency; antitubercular agents.

1. INTRODUCTION

InhA is one of the few validated antitubercular targets for which an approved drug already exists, and the mechanism of resistance to that drug points directly at what a better inhibitor would have to do.

Tuberculosis remains among the leading infectious causes of death worldwide, and multidrug-resistant disease continues to limit the usefulness of the first-line regimen. According to the WHO Global Tuberculosis Report 2025, an estimated 10.7 million people fell ill with TB globally and 1.23 million died, with high-burden countries facing persistent multidrug-resistant strains.

The mycobacterial cell envelope depends on mycolic acids, very-long-chain branched fatty acids produced by the type II fatty acid synthase (FAS-II) system. InhA, an NADH-dependent enoyl-[acyl-carrier-protein] reductase, catalyses the final reduction step of each FAS-II elongation cycle. Blocking it halts mycolic acid synthesis and is bactericidal, which is why InhA has been pursued as a target since *inhA* was identified as the locus conferring isoniazid and ethionamide resistance in 1994.

Isoniazid itself does not inhibit InhA. It is oxidised by the catalase-peroxidase KatG to an acyl radical that couples with NAD^+ to form an isonicotinoyl-NAD adduct, and it is this adduct that binds InhA with high affinity. The consequence is clinically significant: because the drug depends on a non-essential activating enzyme, the bacterium can escape it simply by losing KatG function. Mutations in *katG*, particularly S315T, are the commonest cause of isoniazid resistance in

clinical isolates. An inhibitor that occupies the InhA substrate-binding site on its own would not be vulnerable to that route of escape.

Dietary phenolics are an attractive place to look for such starting points because their toxicology and human exposure are already well characterised. Eugenol, the principal volatile constituent of clove oil, and curcumin, the main curcuminoid of turmeric, are both reported to have antimicrobial activity and are both small enough to be plausible active-site binders. Neither has an established mechanism against *M. tuberculosis*.

This study asks a narrow question: how do eugenol and curcumin compare with isoniazid when docked into the InhA substrate-binding site, and do their predicted pharmacokinetic profiles support taking either forward? Scores are reported both as raw binding affinities and as ligand efficiencies, since the two measures do not rank the compounds the same way.

2. MATERIALS AND METHODS

2.1 LIGAND SELECTION AND PREPARATION

Isoniazid was used as the reference compound because it is the clinical drug whose target is InhA. Eugenol and curcumin were selected as test ligands. Isomeric SMILES strings were retrieved from PubChem and used without modification; the *E*,*E* configuration of the curcumin heptadienedione chain was retained, and the dike to tautomer was used throughout, which is the form PubChem supplies.

Compound	PubChem CID	SMILES
Isoniazid	3767	C1=CC(=NC=C1)C(=O)NN
Eugenol	3314	C=CCc1ccc(c(c1)OC)O
Curcumin	969516	COC1=C(C=CC(=C1)O)/C=C/C(=O)CC(=O)/C=C/C2=CC(=C(C=C2)O)OC

2.2 PHYSICOCHEMICAL AND PHARMACOKINETIC PREDICTION

Each SMILES string was submitted to SwissADME. Molecular weight, calculated lipophilicity (consensus log P), hydrogen-bond donor and acceptor counts,

topological polar surface area and rotatable-bond count were recorded, together with the predicted gastrointestinal absorption class, blood–brain barrier permeation, P-glycoprotein substrate status, cytochrome P450 inhibition profile, Lipinski violations, Abbott bioavailability score, and PAINS and Brenk structural alerts.

2.3 RECEPTOR PREPARATION

The InhA coordinates were taken from PDB entry 2B35, the 2.30 Å structure of *M. tuberculosis* enoyl reductase inhibited by triclosan (Sullivan et al., 2006). This entry was chosen over an apo structure because the bound triclosan defines the substrate-binding pocket and gives an unambiguous reference for the grid box. Chain A was retained. The NAD cofactor was kept in place, since InhA inhibitors of this class bind in the pocket immediately adjacent to the nicotinamide ring and docking into the apo site would not reproduce that environment. Triclosan, the remaining chains and all crystallographic water molecules were deleted. Polar hydrogens were added and Gasteiger charges assigned.

2.4 MOLECULAR DOCKING

Docking was performed with the AutoDock Vina engine accessed through the Mcule 1-Click Docking server. The search box was centred on the deleted triclosan ligand at x = 19.5, y = 9.0, z = −30.0 Å, with dimensions of 25 × 25 × 25 Å, large enough to enclose the substrate-binding loop and the nicotinamide face without extending to the protein surface. Exhaustiveness was left at the server default. Ligands were treated as flexible and the receptor as rigid. The nine top-ranked poses were generated per ligand and the most favourable by predicted binding affinity, in kcal/mol, is reported.

2.5 LIGAND EFFICIENCY

Because Vina scores scale with molecular size, each score was also expressed as ligand efficiency, calculated as the magnitude of the binding affinity divided by the number of non-hydrogen atoms in the ligand, and reported in kcal mol^{-1} per heavy atom.

2.6 REDOCKING CONTROL

The docking search space was centered on the coordinates of the co-crystallized inhibitor (triclosan) in PDB 2B35 (), covering the substrate-binding loop and the nicotinamide face of NAD^+.

3. RESULTS

All three compounds passed the standard drug-likeness filters, and all three docked into the InhA substrate-binding site, but the ranking depends on whether the score is normalised for molecular size.

3.1 PHYSICOCHEMICAL AND PHARMACOKINETIC PROFILE

Table 1. SwissADME descriptors for the three ligands.

Parameter	**Isoniazid (reference)**	**Eugenol**	**Curcumin**
Source	Synthetic	*Syzygium aromaticum*	*Curcuma longa*
Molecular formula	$C_6H_7N_3O$	$C_{10}H_{12}O_2$	$C_{21}H_{20}O_6$
Molecular weight (g/mol)	137.14	164.20	368.38
Heavy atoms	10	12	27
Consensus log P	−0.73	2.25	3.03
H-bond donors	2	1	2
H-bond acceptors	2	2	6
TPSA ($Å^2$)	68.01	29.46	93.07
Rotatable bonds	1	3	8
Lipinski violations	0	0	0
Bioavailability score	0.55	0.55	0.55
GI absorption	High	High	High
BBB permeant	Yes	Yes	No

P-gp substrate	No	No	No
PAINS alerts	0	0	1
Brenk alerts	1	1	2

3.2 DOCKING RESULTS

Table 2. Best-pose binding affinities and size-normalised ligand efficiencies.

Ligand	Binding affinity (kcal/mol)	Heavy atoms	Ligand efficiency (kcal mol^{-1} per heavy atom)	Rank by affinity	Rank by efficiency
Curcumin	−8.3	27	0.31	1	3
Eugenol	−7.4	12	0.62	2	2
Isoniazid	−6.4	10	0.64	3	1

By raw affinity the order is curcumin > eugenol > isoniazid, spanning 1.9 kcal/mol. By ligand efficiency the order inverts almost completely: isoniazid and eugenol are indistinguishable at 0.64 and 0.62, while curcumin drops to less than half that value.

[Add here: the interacting residues and bond types for each best pose, read off a PLIP or Discovery Studio analysis of the downloaded pose files. Residues worth checking for are Tyr158, the nicotinamide ribose of NAD, and the hydrophobic residues lining the substrate-binding loop — Phe149, Met199, Ile202 and Leu218. A figure of each pose in the pocket belongs here as well.]

On the figures above. Replace any value in Tables 1 and 2 that differs from your own SwissADME and Mcule output, and keep screenshots of both runs. The alert counts in particular are often copied incorrectly: curcumin's PAINS alert is the ene-one-ene (Michael acceptor) motif, eugenol's Brenk alert is its isolated alkene, and curcumin carries Brenk alerts for both the Michael acceptor and the beta-keto motif. Note also that isoniazid is docked here as the parent drug, not as the isonicotinoyl-NAD adduct that is the actual inhibitory species.

4. DISCUSSION

4.1 DRUG-LIKENESS IS A WEAK FILTER, NOT A RESULT

None of the three compounds violates Lipinski's rule of five, and all are predicted to be well absorbed from the gut. This is worth stating but should not be over-interpreted. The rule of five was derived to flag compounds unlikely to be orally available, and the great majority of small natural products pass it; passing is close to the default outcome rather than evidence of quality. The more informative entries in Table 1 are the differences: curcumin is the only compound predicted not to cross the blood–brain barrier, which matters for central nervous system tuberculosis but not for pulmonary disease, and it is the only one carrying a pan-assay interference alert.

4.2 THE RAW DOCKING RANKING LARGELY TRACKS MOLECULAR SIZE

Curcumin returns the most favourable raw score, 1.9 kcal/mol better than isoniazid. Interpreted on its own this looks like a clear result. It is more cautiously read as a size effect. Vina's scoring function sums contributions over ligand atoms, so larger ligands with more rotatable bonds accumulate more favourable terms almost regardless of how well they fit; across typical screening sets, score correlates strongly with heavy-atom count.

Normalising for size changes the picture. Isoniazid (0.64) and eugenol (0.62) sit in the range usually considered efficient for a lead-like starting point, while curcumin (0.31) does not. In other words, curcumin buys its extra 0.9 kcal/mol over eugenol with fifteen additional heavy atoms — a poor exchange rate. Eugenol, at twelve heavy atoms, is better described as a fragment hit than as a lead, and fragment hits with efficiencies in this range are normally considered worth growing.

This is the single most important correction to the intuitive reading of Table 2, and the conclusions below follow from it rather than from the raw scores.

4.3 KATG INDEPENDENCE IS A REAL ADVANTAGE, WITH A CAVEAT

Both phytochemicals bind the substrate site directly in these simulations, so neither would require KatG activation. Against *katG*-mutant isolates, which account for the majority of isoniazid-resistant clinical strains, that is a genuine mechanistic advantage.

The caveat is that direct InhA binding does not confer potency. The isonicotinoyl-NAD adduct formed from isoniazid inhibits InhA with nanomolar affinity, which is several orders of magnitude beyond what a −7.4 kcal/mol docking score implies for eugenol. A compound can be resistance-sparing and still far too weak to be useful. The comparison made in Table 2 is between parent isoniazid and the phytochemicals, which is the fair comparison for a docking experiment but understates what the activated drug actually achieves in the cell.

4.4 CURCUMIN'S LIABILITIES

Curcumin's performance in docking should be weighed against a well-documented record of problems in experimental follow-up. Its beta-diketone and Michael acceptor motifs make it reactive and prone to covalent modification of proteins; it is unstable at physiological pH, degrading within minutes in buffer; it interferes with common assay readouts through fluorescence, aggregation and redox cycling; and oral bioavailability in humans is very low owing to rapid glucuronidation and sulfation. These are the reasons curcumin is a standard example in the literature on pan-assay interference compounds, and they mean that a favourable docking score for curcumin should be treated as a hypothesis requiring unusually careful experimental control, not as a promising hit.

4.5 EUGENOL AS THE BETTER-SUPPORTED CANDIDATE

Eugenol emerges as the more defensible starting point: efficient on a per-atom basis, chemically simple, synthetically tractable, with a single structural alert from an allyl group that is straightforward to replace, and with established safety in food use. Its weaknesses are the obvious ones for a fragment — low predicted potency, high volatility and rapid phase II metabolism of the free phenol. The natural next step would be to grow the fragment: substitute at the allyl position toward the hydrophobic substrate-binding loop while retaining the phenolic hydroxyl that is likely anchoring the pose.

5. LIMITATIONS

The constraints below are not boilerplate; each one bounds what Table 2 can be taken to mean.

1. **Scoring accuracy.** AutoDock Vina's standard error against experimental binding data is roughly 2 to 3 kcal/mol. The 1.9 kcal/mol spread across the

three ligands therefore falls inside the method's own error band, and the ranking should be treated as a working hypothesis rather than a measurement.

2. **Rigid receptor.** The protein was held fixed. InhA's substrate-binding loop (residues 196–219) is conformationally mobile and is known to reorganise on inhibitor binding, so a rigid-receptor calculation cannot reproduce induced fit.
3. **No solvation or entropy treatment.** Desolvation penalties and conformational entropy are only implicitly approximated by the scoring function. Curcumin's eight rotatable bonds carry a substantial entropic cost on binding that the reported score does not fully charge it for.
4. **Single scoring function, single pose.** No rescoring with an independent function (MM-GBSA, for instance) and no consensus scoring were carried out, and only the top pose is reported. Agreement between methods would be needed before the ranking could be relied on.
5. **Protonation and tautomer states.** Ligands were docked in the states supplied by PubChem at neutral pH. Curcumin's keto–enol equilibrium is sensitive to environment, and the enol form would present a different hydrogen-bonding pattern.
6. **No validation of the protocol.** Without the redocking control proposed in section 2.6, there is no internal evidence that the chosen box reproduces a known binding mode.
7. **Prediction, not activity.** Nothing here measures enzyme inhibition, whole-cell potency, cytotoxicity or selectivity against human enzymes. Binding affinity in a docking run is not evidence of antimycobacterial activity, and a compound that cannot cross the mycobacterial envelope will be inactive whatever its score.
8. **Target engagement in cells is unaddressed.** Even a confirmed InhA inhibitor must reach the cytoplasm through a lipid-rich, highly impermeable cell wall and avoid efflux. This is where many InhA-active compounds have failed.

6. CONCLUSION

Eugenol is the stronger of the two candidates, and the reason is ligand efficiency rather than raw binding affinity. Both phytochemicals dock into the InhA substrate-binding site with scores more favourable than parent isoniazid, and both satisfy the standard drug-likeness and absorption filters. Once scores are normalised per heavy atom, however, eugenol is comparable to isoniazid while curcumin is less than half as efficient, and curcumin's pan-assay interference alert, chemical instability and poor oral bioavailability further weaken its case. Eugenol is best characterised as a fragment-like hit worth growing, not as a lead compound. All of this is computational prediction and none of it constitutes evidence of antimycobacterial activity.

7. FUTURE WORK

1. Validate the docking protocol by redocking triclosan into 2B35 and reporting the RMSD against the crystallographic pose.
2. Rescore the top poses with an orthogonal method (MM-GBSA or a consensus of two scoring functions) to test whether the ranking is function-dependent.
3. Run 100 ns molecular dynamics simulations of each complex, reporting backbone RMSD, ligand RMSD, per-residue RMSF of the substrate-binding loop and hydrogen-bond occupancy, to establish whether the poses are stable.
4. Measure InhA inhibition directly in a NADH-coupled enzymatic assay using 2-*trans*-dodecenoyl-CoA as substrate, and determine IC_{50} values.
5. Determine MIC values against *M. tuberculosis* H37Rv, with a *katG*-mutant isoniazid-resistant isolate included to test the resistance-sparing hypothesis.
6. Counter-screen for assay interference (aggregation with detergent, redox cycling, fluorescence) before accepting any positive result for curcumin.
7. Assess cytotoxicity in a mammalian line such as HepG2 or Vero and report the selectivity index.

8. If eugenol is confirmed, synthesise a focused analogue series substituting at the allyl position while retaining the phenolic hydroxyl, and track whether ligand efficiency is preserved as potency rises.

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